

Surrogate Modeling in Statistical Learning:

Covariance Functions

Prateek Raj (BT23CS021)

Indian Institute of Technology Madras

Research Intern under the guidance of:

Prof. Neelesh Shankar Upadhye

Department of Mathematics

July 2026

1. Preliminaries & Definitions

- ▶ Role of Covariance Functions
- ▶ Stationarity vs. Isotropy
- ▶ Positive Semidefiniteness (PSD)
- ▶ Mean Square Continuity & Differentiability
- ▶ Bochner's Theorem & Spectral Density

2. Stationary Covariance Functions

- ▶ RBF & γ -Exponential Kernels
- ▶ Matérn Class & Half-Integer Simplifications
- ▶ Rational Quadratic Covariance

3. Non-Stationary Covariance Functions

- ▶ Dot Product Covariance Functions
- ▶ Other Non-Stationary Kernels
- ▶ Making New Kernels from Old

4. Eigenfunction Analysis of Kernels

- ▶ Eigenvalues and Eigenfunctions
- ▶ Mercer's Theorem
- ▶ Analytical & Numerical Approximations

The Core Concept: Covariance defines Similarity

- ▶ **What exactly is a Kernel?** A covariance (kernel) function defines the covariance between the function values at any two points x_i and x_j :

$$k(x_i, x_j) = \text{Cov}(f(x_i), f(x_j))$$

- ▶ **Measuring Relationship:** It tells us how strongly the function values $f(x_i)$ and $f(x_j)$ are related:
 - **Positive Covariance (> 0):** As one function value increases, the other tends to increase.
 - **Negative Covariance (< 0):** As one function value increases, the other tends to decrease.
 - **Zero Covariance ($= 0$):** The function values are not correlated.
- ▶ **The Heart of GP Modeling:** Since GP predictions depend entirely on the covariance function k , kernel design is crucial. It encodes assumptions about the functions, determining properties such as smoothness and periodicity.

Stationarity, Isotropy, and Dot Product Kernels

How does the kernel respond to translation and rotation in input space?

Stationary & Isotropic (Distance-Based)

- **Stationary:** Depends only on the difference vector $\tau = x - x'$ (translation invariant):

$$k(x, x') = k(\tau)$$

Sample functions exhibit homogeneous behavior across the domain.

- **Isotropic:** Depends only on Euclidean distance $r = \|x - x'\|$ (translation and rotation invariant):

$$k(x, x') = k(r)$$

Perfect for fields where close points behave similarly (e.g., temperature, air quality).

Non-Stationary & Dot Product

- **Non-Stationary:** Covariance depends on absolute input coordinates rather than just relative displacement (e.g., spatial models with boundary effects).
- **Dot Product:** Depends only on the inner product $x \cdot x'$ (invariant under orthogonal transformations, but non-stationary):

$$k(x, x') = g(x \cdot x')$$

for some scalar function g . Examples include linear and polynomial kernels.

Mathematical Validity: Positive Semidefiniteness (PSD)

Can we choose any function as a kernel? No, it must satisfy a strict condition: Positive Semidefiniteness (PSD).

Intuition: Variance Can't Be Negative

Let $\mathbf{f}$ be a random vector of GP outputs. Consider any linear combination $\mathbf{z} = \mathbf{a}^T \mathbf{f}$.

$$\text{Var}(\mathbf{z}) = \mathbf{a}^T \mathbf{K} \mathbf{a}$$

Since variance must be non-negative ($\text{Var}(\mathbf{z}) \geq 0$), we require the quadratic form condition:

$$\mathbf{a}^T \mathbf{K} \mathbf{a} \geq 0 \quad \forall \mathbf{a} \in \mathbb{R}^N$$

Any function violating this cannot represent a valid covariance function.

Gram Matrix & Continuous Analogue

- **Gram Matrix:** The covariance matrix $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$ is a Gram matrix, which must be positive semidefinite for any finite set of inputs.
- **Continuous Analogue:** In the continuous limit, the covariance operator must be non-negative:

$$\iint k(\mathbf{x}, \mathbf{x}') f(\mathbf{x}) f(\mathbf{x}') d\mathbf{x} d\mathbf{x}' \geq 0$$

for every square-integrable function $f(\mathbf{x}) \in L^2(\mathcal{X})$, ensuring the covariance operator never produces negative variance.

Shift the question: *What frequencies make up the function?*

Spatial Domain $\leftrightarrow$ Frequency Domain

- ▶ **Smooth Functions:** Contain mainly low frequencies.
- ▶ **Noisy/Rough Functions:** Contain high frequencies.
- ▶ **Spectral Density $S(s)$:** Describes how much power is present at frequency s .
- ▶ **PSD Dual:** Non-negativity in the frequency domain ($S(s) \geq 0$, representing a non-negative power spectrum) is equivalent to positive semidefiniteness in the spatial domain.

Fourier Transform Pairs (Wiener-Khintchine)

$k(\tau)$ and $S(s)$ contain the exact same information:

$$k(\tau) = \int S(s) e^{2\pi i s \cdot \tau} ds$$

$$S(s) = \int k(\tau) e^{-2\pi i s \cdot \tau} d\tau$$

Smoothness Rule: The smoothness of GP sample paths depends on how fast $S(s)$ decays at high frequencies.

Path Properties: Mean Square Continuity & Differentiability

GP path behavior is determined by the local behavior of the kernel near the origin ($\tau \rightarrow 0$).

Mean Square (MS) Continuity

Intuition: Nearby inputs produce identical outputs in the mean-square limit:

$$\lim_{h \rightarrow 0} \mathbb{E} \left[(f(x+h) - f(x))^2 \right] = 0$$

Derivation (for zero-mean stationary processes):

$$\begin{aligned} \mathbb{E} \left[(f(x+h) - f(x))^2 \right] &= \mathbb{E}[f(x+h)^2] + \mathbb{E}[f(x)^2] \\ &\quad - 2\mathbb{E}[f(x+h)f(x)] \\ &= k(0) + k(0) - 2k(h) \\ &= 2[k(0) - k(h)] \end{aligned}$$

As $h \rightarrow 0$, we require $\lim_{h \rightarrow 0} k(h) = k(0)$.

Result: A GP is MS continuous iff its kernel is continuous at the origin.

Mean Square (MS) Differentiability

Intuition: The derivative $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ exists in mean square.

The differentiability of GP paths depends entirely on the differentiability of the kernel.

Theorem: A GP is MS differentiable iff the mixed partial derivative exists and remains finite at $x = x'$:

$$\frac{\partial^2 k(x, x')}{\partial x \partial x'} \text{ exists and is finite.}$$

Upcrossings, Curvature, and Length Scale

How do we quantify "wiggleness"? We count how often a zero-mean, stationary, MS-differentiable GP crosses a threshold level u from below (upcrossings).

Rice's Formula & Curvature

Expected upcrossings in a unit interval:

$$\mathbb{E}[N_u] = \frac{1}{2\pi} \sqrt{\frac{-k''(0)}{k(0)}} \exp\left(-\frac{u^2}{2k(0)}\right)$$

The term $-k''(0)$ is the curvature at the origin, which equals the derivative variance $\text{Var}(f'(x))$:

- ▶ **Large curvature:** Kernel bends sharply $\rightarrow$ covariance drops rapidly $\rightarrow$ function oscillates frequently (many upcrossings).
- ▶ **Small curvature:** Kernel bends gently $\rightarrow$ covariance drops slowly $\rightarrow$ function oscillates slowly (few upcrossings).

The Length Scale Connection

For the Squared Exponential (SE) kernel:

$$k(\tau) = \sigma_f^2 \exp\left(-\frac{\tau^2}{2\ell^2}\right) \Rightarrow -k''(0) = \frac{\sigma_f^2}{\ell^2} \propto \frac{1}{\ell^2}$$

- ▶ **Short length scale $\ell \downarrow$:** Curvature $-k''(0) \uparrow \rightarrow$ many upcrossings $\rightarrow$ very wiggly/rough path.
- ▶ **Long length scale $\ell \uparrow$:** Curvature $-k''(0) \downarrow \rightarrow$ few upcrossings $\rightarrow$ smooth path.

$$\ell \rightarrow k''(0) \rightarrow \text{Crossings} \rightarrow \text{Smoothness}$$

Squared Exponential (SE) Covariance

The Squared Exponential (also known as the RBF or Gaussian kernel) is the most widely-used covariance function:

$$k_{\text{SE}}(r) = \sigma_f^2 \exp\left(-\frac{r^2}{2\ell^2}\right)$$

Advantages & Spectral Decay

- ▶ **Simple & Powerful:** Standard choice for many ML tasks.
- ▶ **Smooth Paths:** Infinite differentiability of the kernel at the origin guarantees infinitely mean-square differentiable sample paths.
- ▶ **Exponential Decay:** In Bochner's view, its spectral density decays exponentially fast:

$$S(s) \propto \exp(-cs^2)$$

Stein's Critique (Stein 1999)

Michael Stein argued that infinite smoothness is unrealistic for physical processes (e.g., temperatures, geology) because the sample paths are analytic, meaning that observations in a small interval determine the entire function history with probability 1 (analytic continuation). He recommended the **Matérn** class.

The γ -Exponential Covariance

This family generalizes both the exponential and squared exponential kernels:

$$k(r) = \exp\left(-\left(\frac{r}{\ell}\right)^\gamma\right) \quad \text{for } 0 < \gamma \leq 2$$

Special Cases

- **Exponential/OU Kernel ($\gamma = 1$):**

$$k(r) = \exp\left(-\frac{r}{\ell}\right)$$

Yields continuous but nowhere differentiable paths.

- **Squared Exponential Kernel ($\gamma = 2$):**

$$k(r) = \exp\left(-\frac{r^2}{\ell^2}\right)$$

Yields infinitely differentiable paths.

Differentiability Limitation

Unlike the Matérn class, which allows continuous control of differentiability, the γ -exponential process is:

- **Not MS differentiable** for any $\gamma < 2$.
- **Infinitely MS differentiable** for $\gamma = 2$.

This lack of intermediate smoothness limits its flexibility.

The Matérn Class: Controllable Smoothness

The Matérn class allows explicit control over the differentiability of the sample paths:

$$k_{\text{Matern}}(r) = \sigma_f^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu} r}{\ell} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu} r}{\ell} \right)$$

where $\nu > 0$ is the smoothness parameter, and K_ν is the modified Bessel function of the second kind of order ν .

- ▶ **Differentiability:** GP paths are m -times MS differentiable if and only if $\nu > m$.
- ▶ **Frequency Interpretation:** The spectral density decays polynomially (power-law):

$$S(s) \propto \left(\frac{2\nu}{\ell^2} + 4\pi^2 s^2 \right)^{-(\nu+D/2)}$$

High frequencies survive longer than in the SE kernel, making Matérn more realistic for physical data (e.g., wind, temperature).

- ▶ **Limit:** As $\nu \rightarrow \infty$, the Matérn kernel converges to the SE kernel.

Matérn: Half-Integer Simplifications

When $\nu = p + 1/2$ (for integer $p \geq 0$), the Bessel function disappears, leaving a product of a polynomial of order p and an exponential decay:

$\nu = 1/2$ (Exponential)

$$k_{1/2}(r) = \sigma_f^2 \exp\left(-\frac{r}{\ell}\right)$$

- ▶ MS continuous but *nowhere* MS differentiable.
- ▶ Corresponds to the **Ornstein-Uhlenbeck (OU)** process (describes Brownian velocity).

$\nu = 3/2$ (Moderate)

$$k_{3/2}(r) = \sigma_f^2 \left(1 + \frac{\sqrt{3}r}{\ell}\right) \exp\left(-\frac{\sqrt{3}r}{\ell}\right)$$

- ▶ *Once* MS differentiable.
- ▶ Ideal for modeling physical fields with moderately abrupt transitions.

$\nu = 5/2$ (Standard)

$$k_{5/2}(r) = \sigma_f^2 \left(1 + \frac{\sqrt{5}r}{\ell} + \frac{5r^2}{3\ell^2}\right) \exp\left(-\frac{\sqrt{5}r}{\ell}\right)$$

- ▶ *Twice* MS differentiable.
- ▶ Standard recommendation for most machine learning tasks.

The Rational Quadratic covariance is equivalent to an infinite mixture of SE kernels with different length-scales:

$$k_{\text{RQ}}(r) = \sigma_f^2 \left(1 + \frac{r^2}{2\alpha\ell^2} \right)^{-\alpha}$$

where $\alpha > 0$ controls the shape of the mixture.

► **Multi-Scale Modeling:**

Useful for processes containing both fast local variations and slow global trends. As $\alpha \rightarrow \infty$, the RQ kernel converges back to the SE kernel:

$$\lim_{\alpha \rightarrow \infty} \left(1 + \frac{r^2}{2\alpha\ell^2} \right)^{-\alpha} = \exp \left(-\frac{r^2}{2\ell^2} \right)$$